

EE Quiz 2

MS101 EE Quiz 2 for regular students

1.

An experiment similar to Expt 5 was carried out using a BO motor with a disk having 8 sectors (alternately white and black) connected to its shaft. A TCRT sensor (LED-photo transistor pair) was used to detect the sectors. If the BO is rotating at 180 RPM what will be the output waveform time period, in ms? Your answer should be precise to the first decimal place.

Marks: 3

Type: FLOAT_TYPE

2. Mark **all the correct options** in the following:

Marks: 2

Type: MULTIPLE_CHOICE

For PID control of a heater,

Options:

- 0) the integral term is needed to make the error zero at the set point.
 - 1) the differential term is needed to react to fast changes in the error.
 - 2) the integral term is needed to react to fast changes in the error.
 - 3) the differential term is needed make the error zero at the set point.
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3. Mark **all the correct options** in the following:

Marks: 2

Type: MULTIPLE_CHOICE

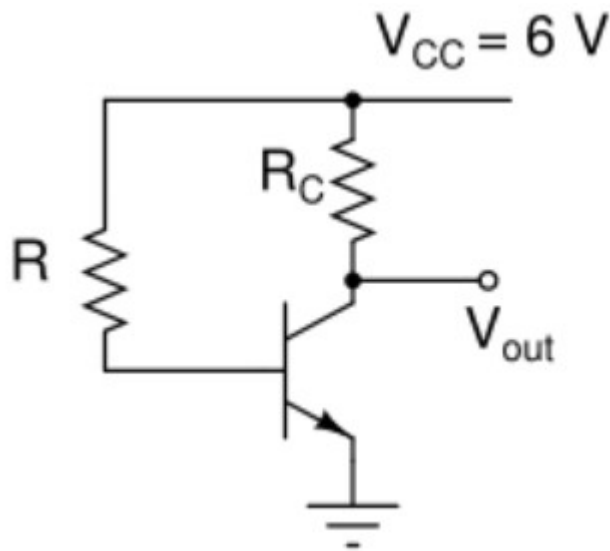
Hysteresis is needed in the ON-OFF control of a heater to

Options:

- 0) make the error zero at the set point.
 - 1) avoid frequent switching of the relay controlling the heater.
 - 2) avoid oscillations in the temperature.
 - 3) make the control faster.
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4.

In the BJT circuit shown below, determine the value of the resistor R in $k\Omega$ if the output voltage V_{out} is 3 V. Given: $\beta = 100$, $R_C = 1 k\Omega$, $V_{BE} = 0.6 V$, $V_{CEsat} = 0.2 V$. Your answer must be precise to the first decimal place.



Marks: 3

Type: FLOAT_TYPE

5.

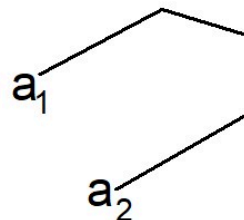
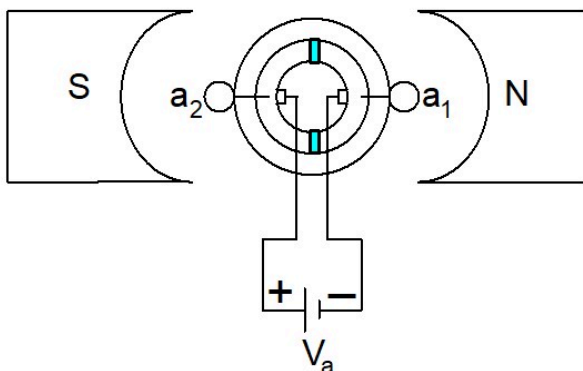
In a BJT circuit $\beta = 80$. If the emitter current I_E in the circuit is 2 mA, what will be the collector current I_C in mA? Your answer should be precise to the second decimal place.

Marks: 2

Type: FLOAT_TYPE

6.

Schematic diagram of a single coil dc motor having a single coil $a_1 - a_2$, is shown in the following figure. Considering the polarity of the applied voltage as marked in the diagram, determine in which direction (clockwise or anticlockwise as viewed from the front) the armature will rotate?



Marks: 1

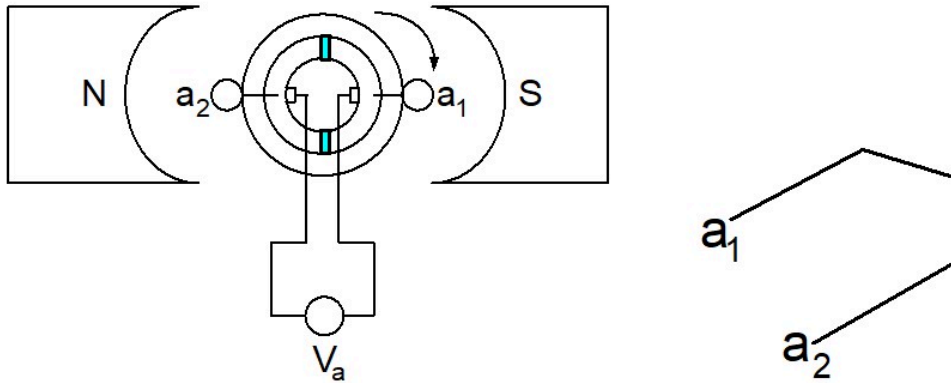
Type: SINGLE_CORRECT_ANSWER

Options:

- 0) clockwise
- 1) anticlockwise

7.

An armature voltage V_a is applied to the dc motor shown below. Polarity of V_a is not provided. It is found that the armature of the motor is rotating in the clockwise direction when viewed from the front as depicted in the figure. Determine whether the polarity of the back emf, $E_{b(a2,a1)}$ (i.e. the potential of the armature terminal, a_2 measured with respect to the terminal a_1) would be positive or negative.



Marks: 1

Type: SINGLE_CORRECT_ANSWER

Options:

- 0) positive
- 1) negative

8. The speed of a BO type dc motor can be reduced by

Marks: 2

Type: SINGLE_CORRECT_ANSWER

Options:

- 0) either increasing the resistance in series with the armature or by increasing the applied armature voltage.
- 1) either increasing the resistance in series with the armature or by decreasing the applied armature voltage.
- 2) either decreasing the resistance in series with the armature or by increasing the applied armature voltage.
- 3) decreasing the resistance in series with the armature while maintaining a constant applied armature voltage.

9. Mark **all the correct ones** out of the following statements for MOSFETS:

Marks: 2

Type: MULTIPLE_CHOICE

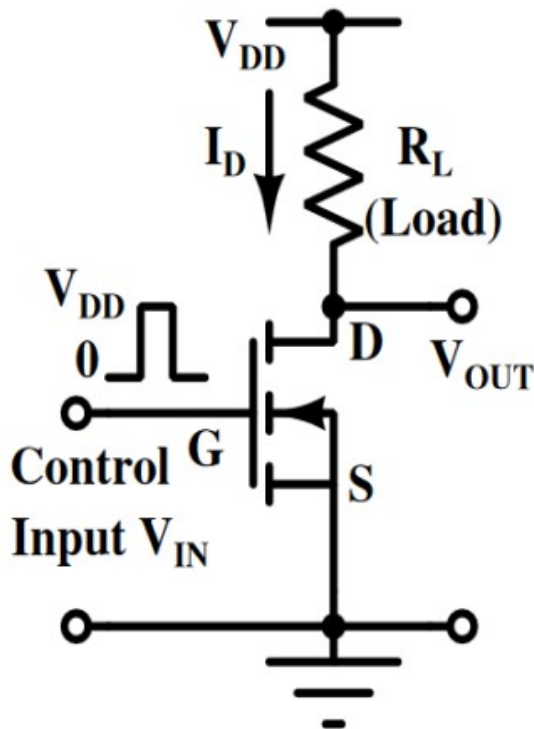
Options:

- 0) A MOSFET is ON when there is a small resistance for current flow between its gate and drain terminals.
- 1) An enhancement-mode MOSFET has no drain-source channel when the gate-to-source voltage is zero ($V_{GS} = 0$).
- 2) A P-channel enhancement-mode MOSFET has a positive threshold voltage ($V_T > 0$).

3) In an N-channel MOSFET, the drain-to-source voltage is positive ($V_{GS} > 0$)

10.

The figure shows a load switched using a MOSFET. Given: Supply voltage $V_{DD} = 10$ V, threshold voltage $V_T = 2$ V, and load $R_L = 1$ k Ω . Mark **all the correct ones** out of the following statements for this circuit:



Marks: 3

Type: MULTIPLE_CHOICE

Options:

0) The load current I_D is zero for $V_{IN} < 2$ V and about 10 mA for 10 V $> V_{IN} > 2$ V.

1) The load current I_D is zero for $V_{IN} < 2$ V and increases from zero to about 10 mA as V_{IN} increases from 2 V to 10 V.

2) The load current I_D is zero for $V_{IN} < 8$ V and increases from zero to about 10 mA as V_{IN} increases from 8 V to 10 V.

3) The current drawn from the control input is zero for 10 V $> V_{IN} > 0$.

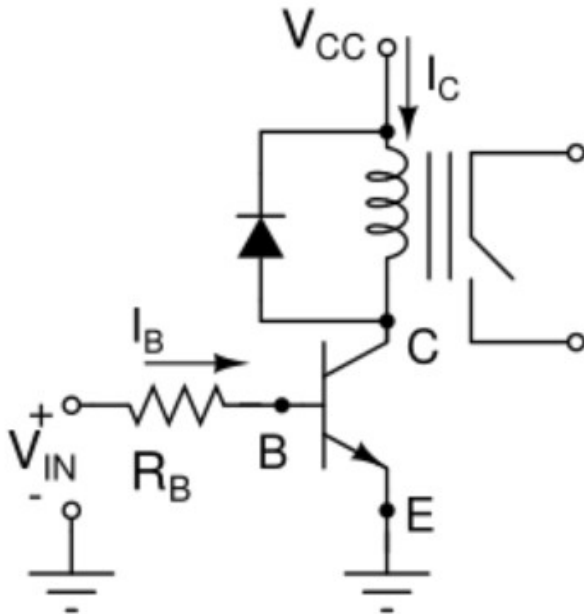
11.

A BJT circuit for driving a relay is shown below.

$V_{cc} = 10$ V, $R_{Coil} = 400$ Ω , $R_B = 10$ k Ω , $\beta = 50$. Minimum current required for the relay to get activated is 16 mA. R_{Coil} is the relay coil resistance.

BJT parameters: Nominal $V_{BE} = 0.6$ to 0.8 V, $V_{CEsat} = 0.2$ V.

If $V_{IN} = 10\text{ V}$, what is the current through the relay in mA? Your answer should be precise to the first decimal place.



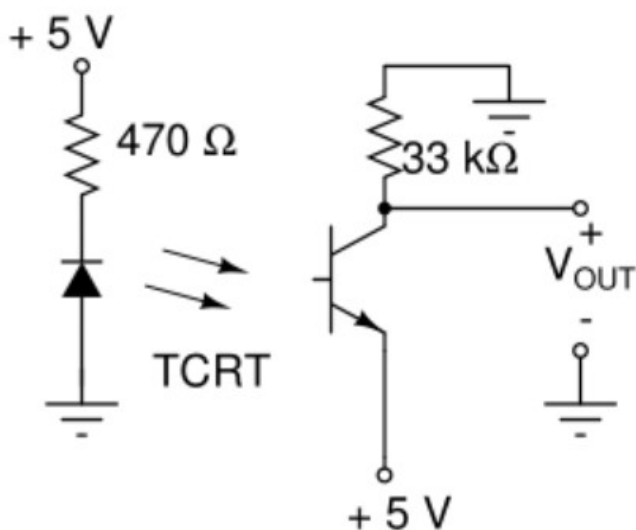
Marks: 2

Type: FLOAT_TYPE

12.

The following circuit diagram was used to conduct an experiment (similar to the one done in Expt 5) using the TCRT (LED-photo transistor pair) sensor for measuring the RPM of a BO motor. Output V_{OUT} was connected to the DSO for observing the time period of the TCRT output.

Out of the following statements mark **all the correct ones** for the circuit shown.



Marks: 2

Type: MULTIPLE_CHOICE

Options:

- 0) Circuit diagram for the LED part of the circuit is correct.
 - 1) Circuit diagram for the LED part of the circuit is wrong.
 - 2) Circuit diagram for the photo transistor part is correct.
 - 3) Circuit diagram for the photo transistor part is wrong.
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